

The empirical recurrence for OEIS A232960, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A232960 records “Empirical recurrence of order 86” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 94$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 88$. Its last coefficient is $c_{86} = 4131221667840000 \neq 0$, so no further tail satisfies anything shorter; any order-86 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A232960 is “Number of $6 \times n$ 0..3 arrays with no element $x(i,j)$ adjacent to value $3-x(i,j)$ horizontally, diagonally or antidiagonally, and top left element zero.”. It has offset 1 and begins

1024, 50421, 5840289, 609853251, 66226131273, 7168373617095, 777971521023633, 84438762415260969, ...

The entry states:

Empirical recurrence of order 86 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 07:46:19, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 94$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 94$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 86: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 86.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 188$ exact terms from $n = 2$ on, it returns order 86 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{86} c_i a(n-i),$$

c_1	148	c_{23}	−8037067187065791058056423	c_{45}	4116441644482429087
c_2	−2358	c_{24}	−1087016222963545499262096653	c_{46}	−5478761434466395474
c_3	−291232	c_{25}	1884268713250776176440835301	c_{47}	12794970092589666978
c_4	7848362	c_{26}	17429118394683622040309417082	c_{48}	90120368514103284313
c_5	186871549	c_{27}	−54617414298915784305568516885	c_{49}	−3774711881852031492
c_6	−7189765941	c_{28}	−188541129867233399120435456466	c_{50}	−11599056356804091775
c_7	−32415036626	c_{29}	935308218780877385665335536094	c_{51}	65206637404977655114
c_8	3003327228438	c_{30}	1219332306527598634100247045573	c_{52}	11708454025597907815
c_9	−11509851978283	c_{31}	−11043006607505777400738265502582	c_{53}	−7927156037206304813
c_{10}	−625891536640949	c_{32}	−1441146432888663614886649773908	c_{54}	−9273287482578363561
c_{11}	5805273937376450	c_{33}	94605776194288662702026552367921	c_{55}	71619541043326442548
c_{12}	62492319492445008	c_{34}	−62542167545045372674432910939984	c_{56}	57542016341665932654
c_{13}	−1001809908171247858	c_{35}	−600141508686600616882839052691915	c_{57}	−4914714518800943703
c_{14}	−2041320705135272343	c_{36}	779563298283961274811769497340030	c_{58}	−2787356088028613459
c_{15}	92441065706906838932	c_{37}	2829876835421223788129810158669205	c_{59}	25836803922988116304
c_{16}	−173668644581441758286	c_{38}	−5442717046547176063192193637954691	c_{60}	10475389349299741024
c_{17}	−5100660470079867532792	c_{39}	−9795003241864281372595786866084297	c_{61}	−1041807409656019617
c_{18}	23686040064413605484232	c_{40}	26405468699109964994289339871362742	c_{62}	−3025441983345659454
c_{19}	170084800330726590602558	c_{41}	23821228821616785883243060728803710	c_{63}	3209040610345030153
c_{20}	−1342645524366466641352211	c_{42}	−94854151337536037202779694567724971	c_{64}	6621889892142197601
c_{21}	−2937777927886856676300837	c_{43}	−34954736420927220706993864791507716	c_{65}	−748492884817434998
c_{22}	46643407166031455359111683	c_{44}	259390172754265881262747604551987796	c_{66}	−107624806237707472

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 88$, and it is the recurrence OEIS A232960 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{86} - c_1x^{86-1} - \dots - c_{86}$. Its constant term is $-c_{86} = -4131221667840000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 86 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 86, so $\deg q = 86$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 86, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 12 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 86, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4131221667840000.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-86 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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